

INDIVIDUAL PROGRESS REPORT

Task 9: Integrated Pipeline, Algorithm Metrics, and Professional Visualizations

Group Topic	Boardie Analytics: A Big Data Approach to Student Boarding House Discovery and Geofenced Presence Pattern Analysis
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Lab No.	Task 9
Git Repo / Colab Link	https://github.com/carlpoot/week-4-biggs-data

Objectives

The main objective of Task 9 is to enhance the earlier Boardie Analytics algorithm project into a more comprehensive analytics pipeline that integrates external tools and generates professional-quality visualizations. The activity emphasizes the preparation of dashboard-ready processed data, exporting algorithm evaluation metrics, producing visual summaries, and organizing all outputs for GitHub submission and documentation.

Tasks Completed

- Examined the requirements of Task 9 and linked the activity to the previous Week 3 Random Forest classification model.
- Developed an integrated analytics pipeline that imports the processed Boardie dataset and generates derived variables for dashboard applications, such as availability category, review quality, amenity level, and listing size.
- Continued the classification implementation by training a Random Forest model to categorize listing price bands using features related to property details, room type, availability, reviews, and listing characteristics.
- Optimized the model through GridSearchCV by tuning parameters including the number of trees, maximum depth, and minimum samples per leaf.
- Produced algorithm evaluation metrics such as accuracy, weighted precision, weighted recall, weighted F1 score, classification report, and confusion matrix.
- Exported processed datasets for integration with external visualization platforms, particularly a dashboard-ready CSV file compatible with Tableau, Power BI, Looker Studio, and spreadsheet-based dashboards.
- Generated professional visualization outputs including price-band distribution, room type by price band, availability by price band, feature importance, confusion matrix, neighbourhood median price, and association-rule lift charts.

- Included a basic association-pattern summary utilizing support, confidence, and lift to align the output with the Chapter 9 topic on association rule mining.
- Created a browser-based HTML dashboard preview to present charts and metrics in a professional format.
- Saved the notebook, Python script, processed CSV files, visualization outputs, README, and individual progress report for GitHub submission.

Tasks to be Completed

- Upload the Task 9 notebook, script, processed CSV outputs, visualization files, and report to the group GitHub repository.
- Update the placeholder GitHub link in the progress report once the repository is finalized and the files are uploaded.
- If requested by the instructor, import the dashboard-ready CSV into Tableau or another external platform and recreate the charts as an interactive dashboard.
- Further improve the model by experimenting with additional algorithms or incorporating more contextual features if the project dataset expands.

Pipeline Summary

Pipeline Stage	Output / Purpose
Processed Dataset	Utilizes the Week 3 processed Boardie analytics dataset as the primary input.
Feature Preparation	Generates model-ready and dashboard-ready variables such as review quality, availability category, and amenity level.
Model Training	Trains a Random Forest Classifier for listing price-band classification.
Algorithm Metrics	Produces accuracy, weighted precision, weighted recall, weighted F1 score, classification report, and confusion matrix outputs.
External Tool Integration	Exports tableau_boardie_dashboard_data.csv and additional CSV summaries for Tableau, Power BI, Looker Studio, or spreadsheet dashboards.
Professional Visualizations	Produces PNG charts and a dashboard_preview.html file for presentation-ready viewing.

Algorithm Metrics Summary

Metric	Value
Accuracy	0.7200
Weighted Precision	0.7278
Weighted Recall	0.7200
Weighted F1 Score	0.7049
Training Rows	750
Testing Rows	250
Best Parameters	{"model__max_depth": 10, "model__min_samples_leaf": 1, "model__n_estimators": 100}

Results and Analysis

Task 9 successfully produced outputs for both model evaluation and professional visualization. A Random Forest Classifier was utilized to categorize listings into different price groups using listing and availability-related features as inputs. After conducting hyperparameter tuning, the model achieved an accuracy score of 0.7200 and a weighted F1 score of 0.7049. These results indicate that the model can provide a reliable baseline for predicting listing price categories, although budget and premium categories remain less frequent compared to mid-range and upper-mid listings.

The exported CSV files demonstrate the integration component of the task. Instead of limiting the results to notebook-based outputs, the generated files can be imported into external tools such as Tableau, Power BI, Looker Studio, or spreadsheet applications for additional analysis and visualization.

The generated visualizations illustrate the findings obtained from the data analysis process. The price-band and room-type charts display the distribution of listings, while the availability and feature-importance diagrams highlight the patterns that influenced the model's predictions. The confusion matrix identifies which predictions were classified correctly or incorrectly. Furthermore, the association-pattern outputs relate to the Chapter 9 discussion on association rule mining, as they present co-occurrence relationships using support, confidence, and lift measurements.

Figure 1. Listing distribution by price band

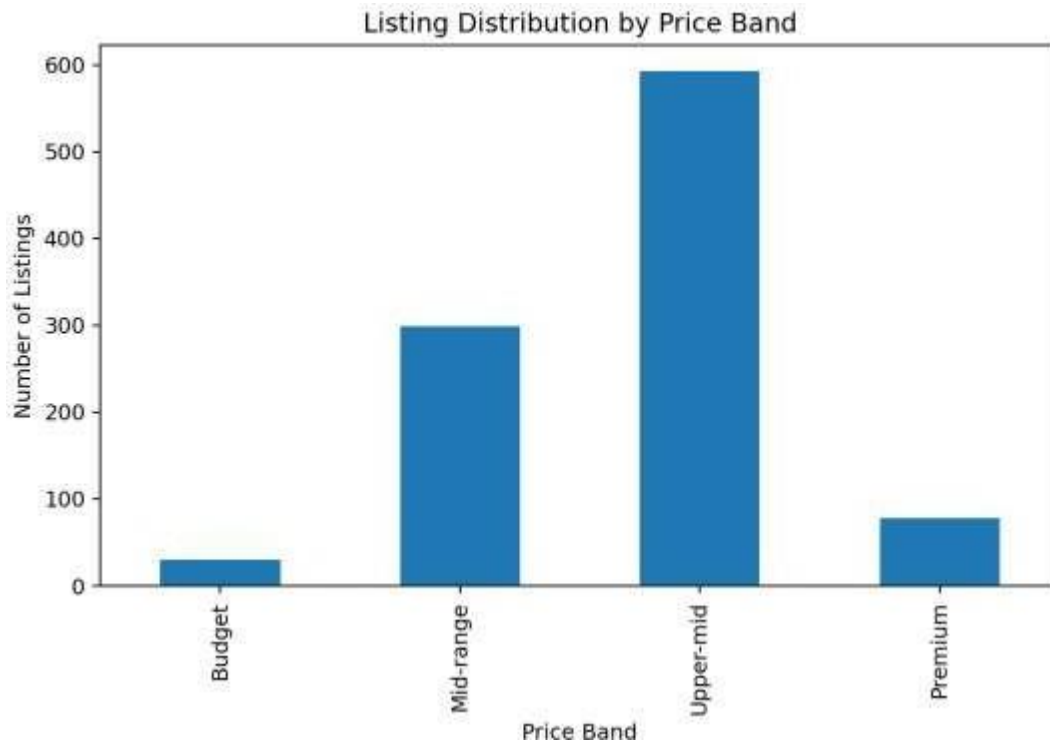


Figure 2. Price band composition by room type

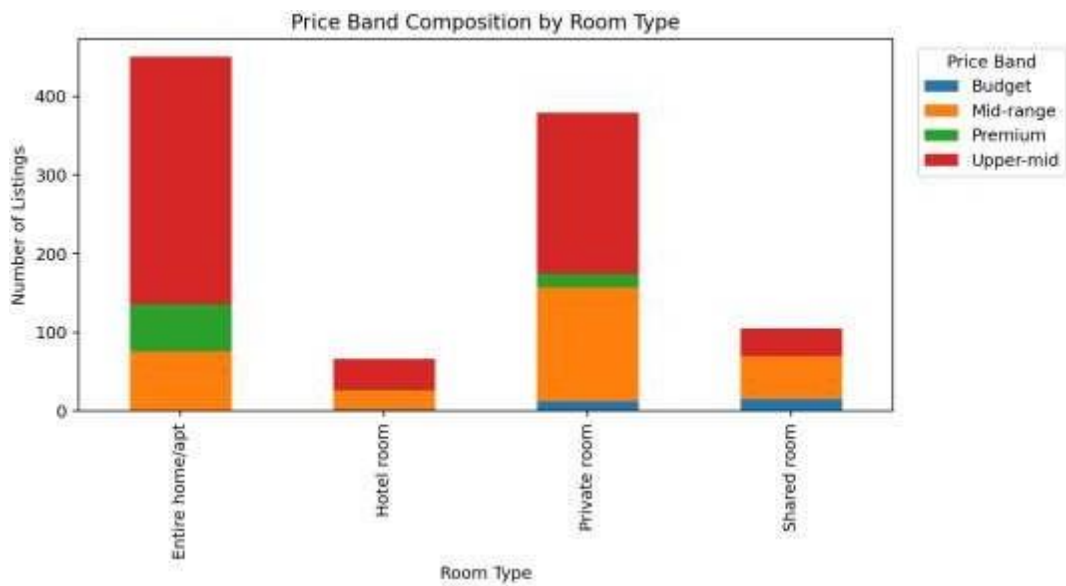


Figure 3. Average availability rate by price band

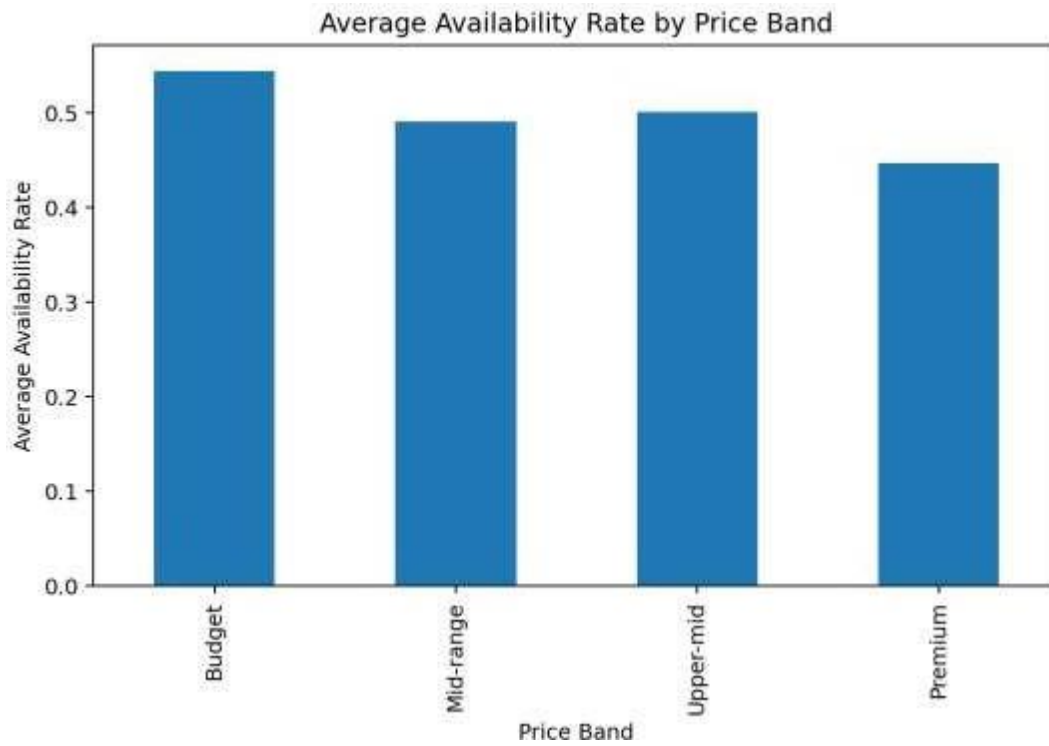


Figure 4. Feature importance for price-band classification

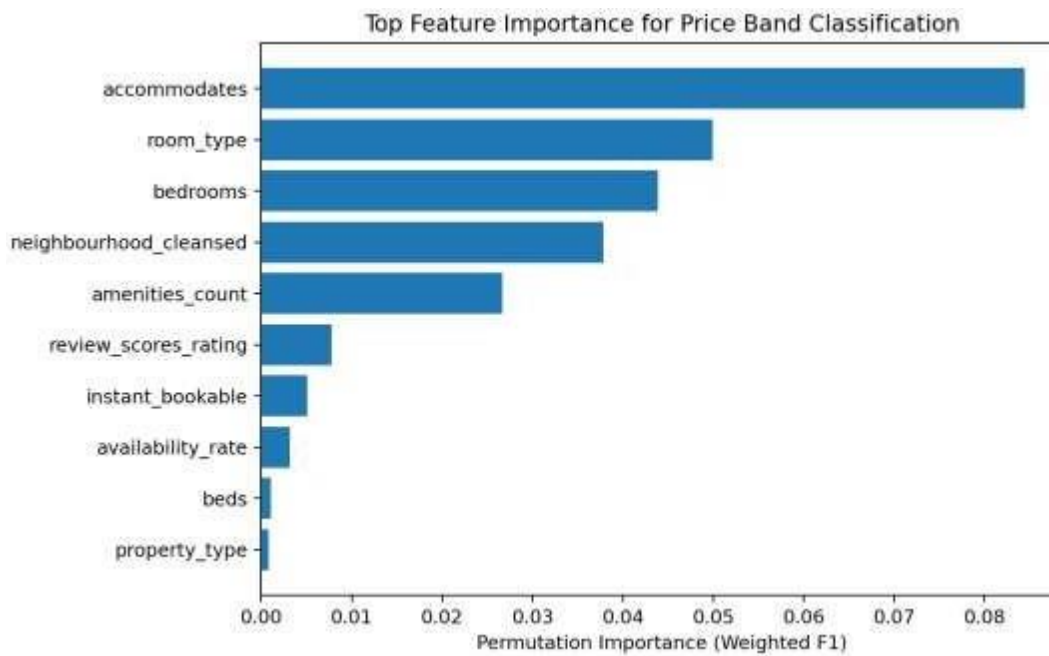


Figure 5. Random Forest confusion matrix

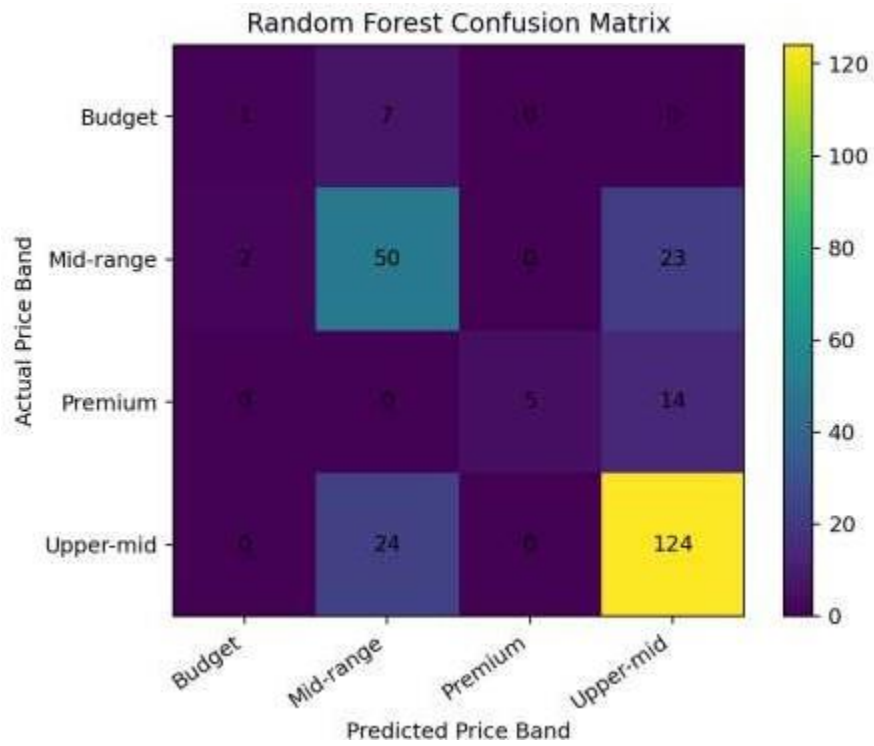


Figure 6. Top areas by median listing price



Challenges and Solutions

Challenge	Solution
The activity requires compatibility with external visualization platforms, although the group has not yet finalized a Tableau dashboard.	The pipeline was designed to export dashboard-compatible CSV datasets along with an HTML dashboard preview, allowing the outputs to be transferred later into Tableau or similar BI applications.
Certain price-band categories appear less often in the dataset, which may reduce the consistency of the classification results.	To address class imbalance, the model implemented stratified train-test splitting together with class-weight adjustments during training.
Producing professional-quality visual outputs requires more advanced presentation compared to ordinary exploratory graphs.	The project generated polished charts containing proper titles, labels, summarized dashboard information, and a browser-based preview intended for presentation and reporting purposes.
Chapter 9 emphasizes association rule mining techniques, while the earlier phase of the project mainly focused on classification algorithms.	A simplified association-pattern analysis using support, confidence, and lift metrics was integrated while maintaining the existing Random Forest classification performance outputs.

Conclusion

Task 9 was successfully completed by transforming the earlier Boardie Analytics model into a more comprehensive analytics pipeline. The project produced optimized classification results, algorithm performance metrics, dashboard-ready CSV files, professional visualizations, an HTML dashboard preview, and updated project documentation. Through this activity, the processed data was effectively prepared for integration with external visualization and business intelligence platforms, demonstrating the transition from analytical coding outputs to presentation-ready reporting tools.